Lesson 5 - Scale and Encode

READ ME FIRST (theory). Then open the practice notebook.

Contents

Lesson 5 - Scale and Encode (and the most important rule)

The model only understands **numbers on a similar scale**. The theory, then `practice.ipynb`.

Key words

Word	Simple meaning
scale	make numbers a similar size
StandardScaler	makes mean = 0 and spread = 1
encode	turn text categories into numbers
OneHotEncoder	makes a 0/1 column for each category
fit	LEARN from the training data
transform	APPLY what was learned
leakage	the model secretly sees test data - bad

1. The most important rule: split first

Split the data into **train** (80%) and **test** (20%) **before** you prepare it. The model learns from train. We keep test hidden to check it honestly.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

Why: if we prepare using the test data too, the model gets secret help. That is **leakage**.

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2. Scale numbers - StandardScaler

Some columns are big (TotalCharges = 8000), some small (tenure = 5). The model thinks the big numbers matter more. We fix this: make every column mean 0 and spread 1.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(X_train[num_cols])                  # LEARN from train
train_scaled = scaler.transform(X_train[num_cols])   # APPLY
test_scaled  = scaler.transform(X_test[num_cols])    # APPLY (same numbers)
```

Why: so big and small columns are treated fairly.

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3. fit vs transform (remember this!)

- **fit** = LEARN (the mean, the median, the categories) from the **training** data.
- **transform** = APPLY what was learned to change the data.
- We **fit on train only**, then **transform** both train and test.

Common mistake: calling `fit` on the test data. That is leakage.

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4. Encode text - OneHotEncoder

The model cannot read words. We turn each category into 0/1 columns.

```
from sklearn.preprocessing import OneHotEncoder
enc = OneHotEncoder(handle_unknown="ignore", sparse_output=False)
```

Example: `Contract = "Two year"` becomes `[0, 0, 1]`.

`handle_unknown="ignore"` = if a NEW category appears later, do not crash.

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5. What is leakage, simply?

Leakage = the model sees information it should not see while learning. The test score looks great, but in real life the model fails. Splitting first and fitting on train only prevents it.

Quick summary

- **Split first** (train/test). Never let the model see test data while learning.
- **StandardScaler** makes numbers a fair size.
- **OneHotEncoder** turns text into 0/1 columns; use `handle_unknown="ignore"`.
- **fit = learn (train only), transform = apply (train and test).**
- Fitting on test data = **leakage** = bad.

Now do this

Open `practice.ipynb`, split + scale + encode the real data, and do the assignment.